

Internal Chatbot for ETH AI CENTER

Enhancing internal collaboration with AI

Data Science Lab group 7, supervised by Imanol Schlag



ETH AI CENTER

Project Overview

Motivation:

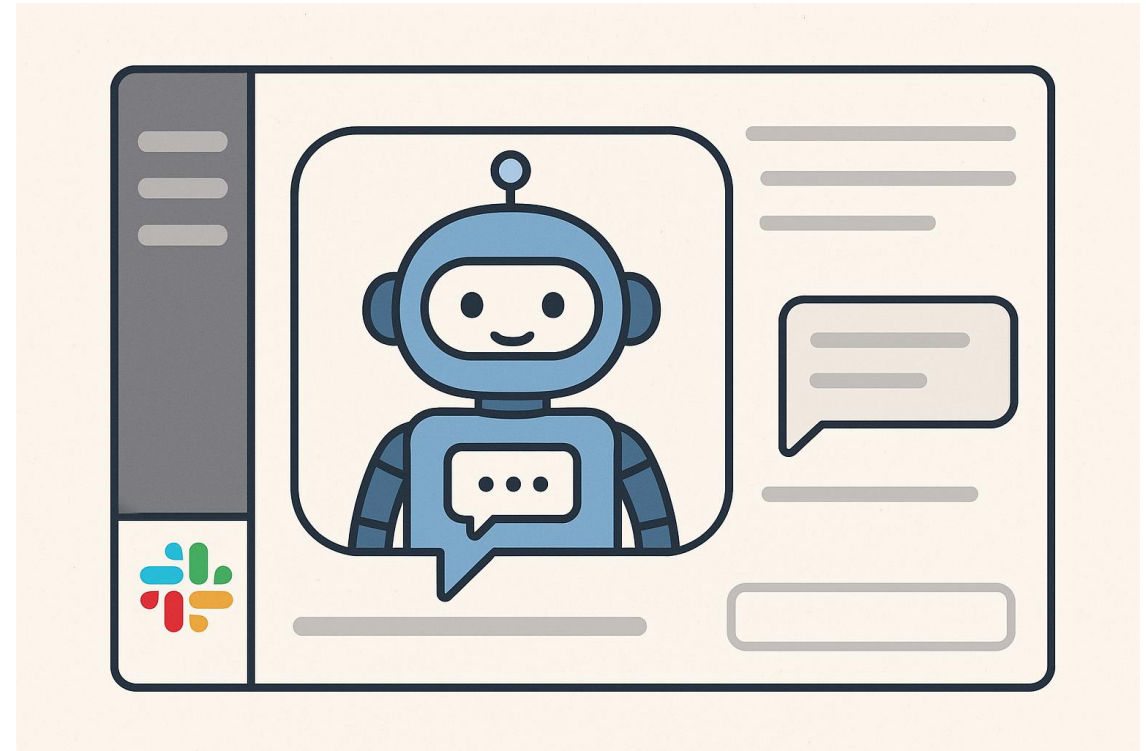
- We often waste time searching through docs or asking around for information

Goal:

- Build a smart internal assistant, integrated in Slack, that can answer questions using the knowledge of the ETH AI Center

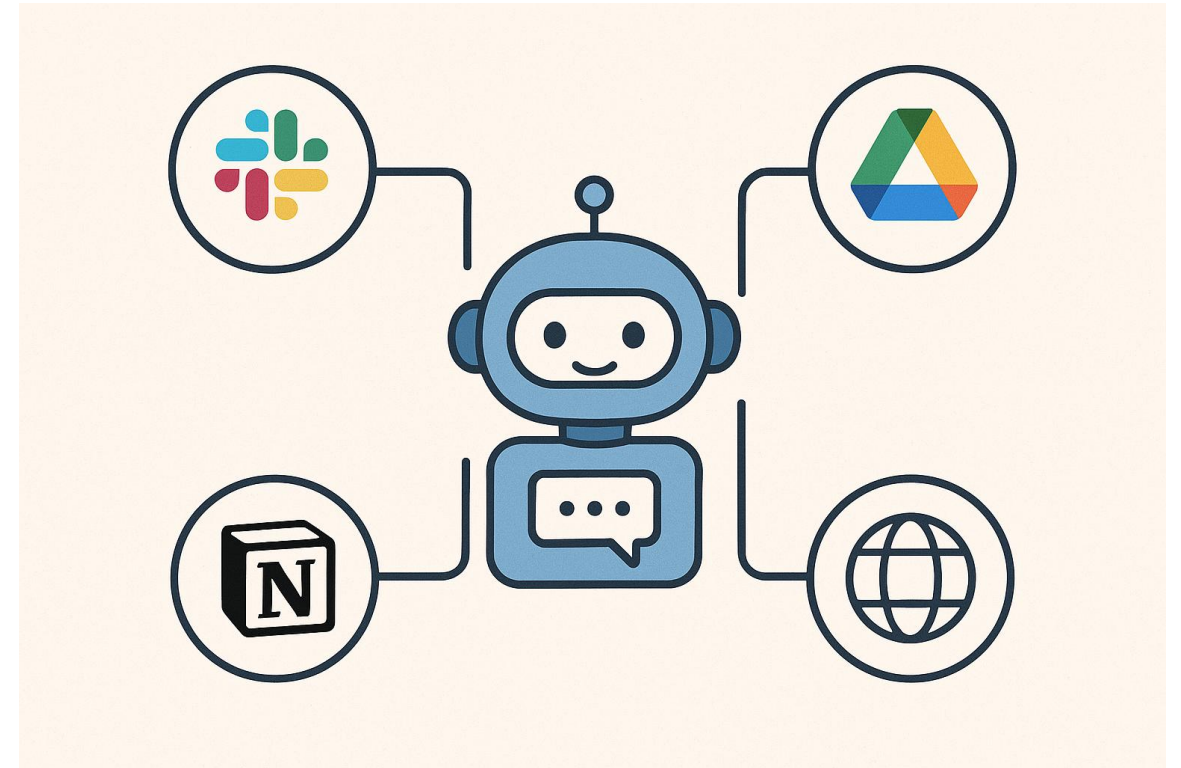
Benefits:

- Save time
- Reduce complicated communication
- Centralise knowledge access



Data Sources

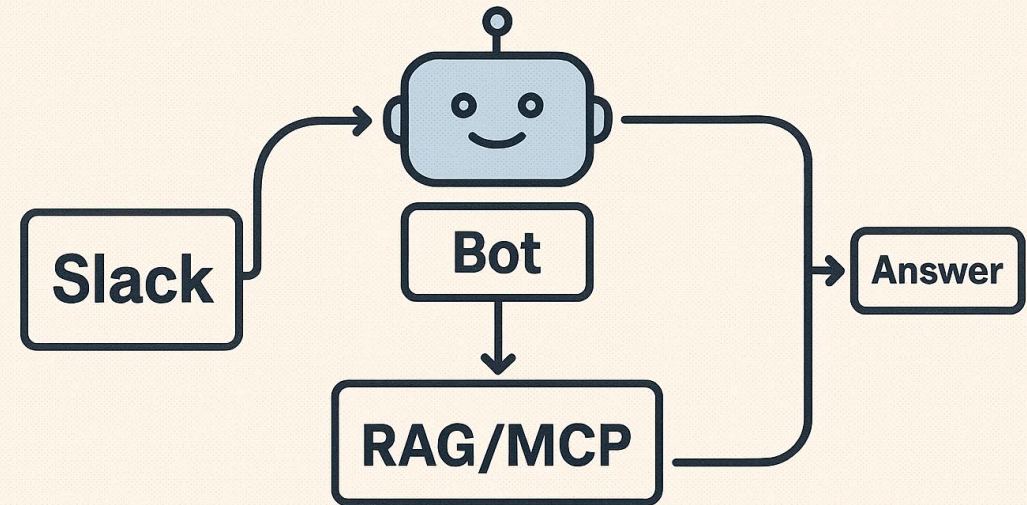
- Slack : messages (pinned/context when bot is invoked)
- Notion : project pages, documentation
- Google Drive : PDFs, Docs, Sheets and most other formats
- Web pages : wiki, lab websites and more (waiting on Flavio's insight)
- Airtable ? Potential, exploratory
- We welcome your ideas for other sources we should consider!



Architecture Overview

How the system works :

- MCP protocol for some data access
- Documents crawled and stored in vector database for some other data sources
- RAG (retrieval-augmented generation) for answering queries

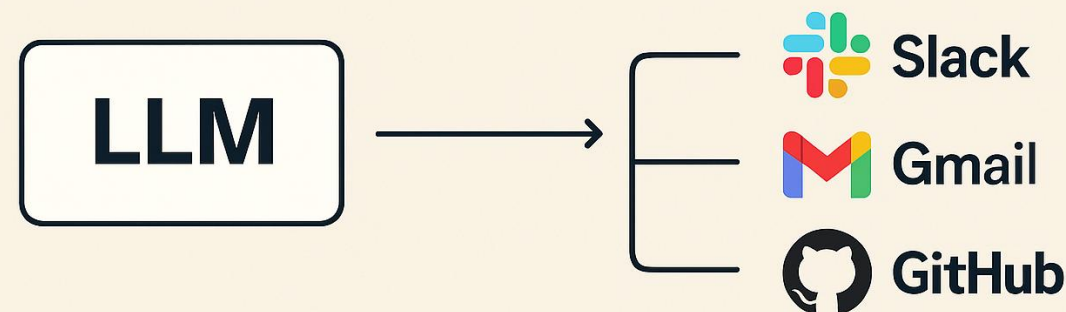


Model Context Protocol (MCP)

- Standardises how applications provide context to LLMs
- A list of pre-built tools that the LLM can directly use
- Follows a client-server architecture
- Best practices for securing data within the infrastructure (locally run)

Model Context Protocol

A protocol that allows a language model to utilize additional context from external data sources.

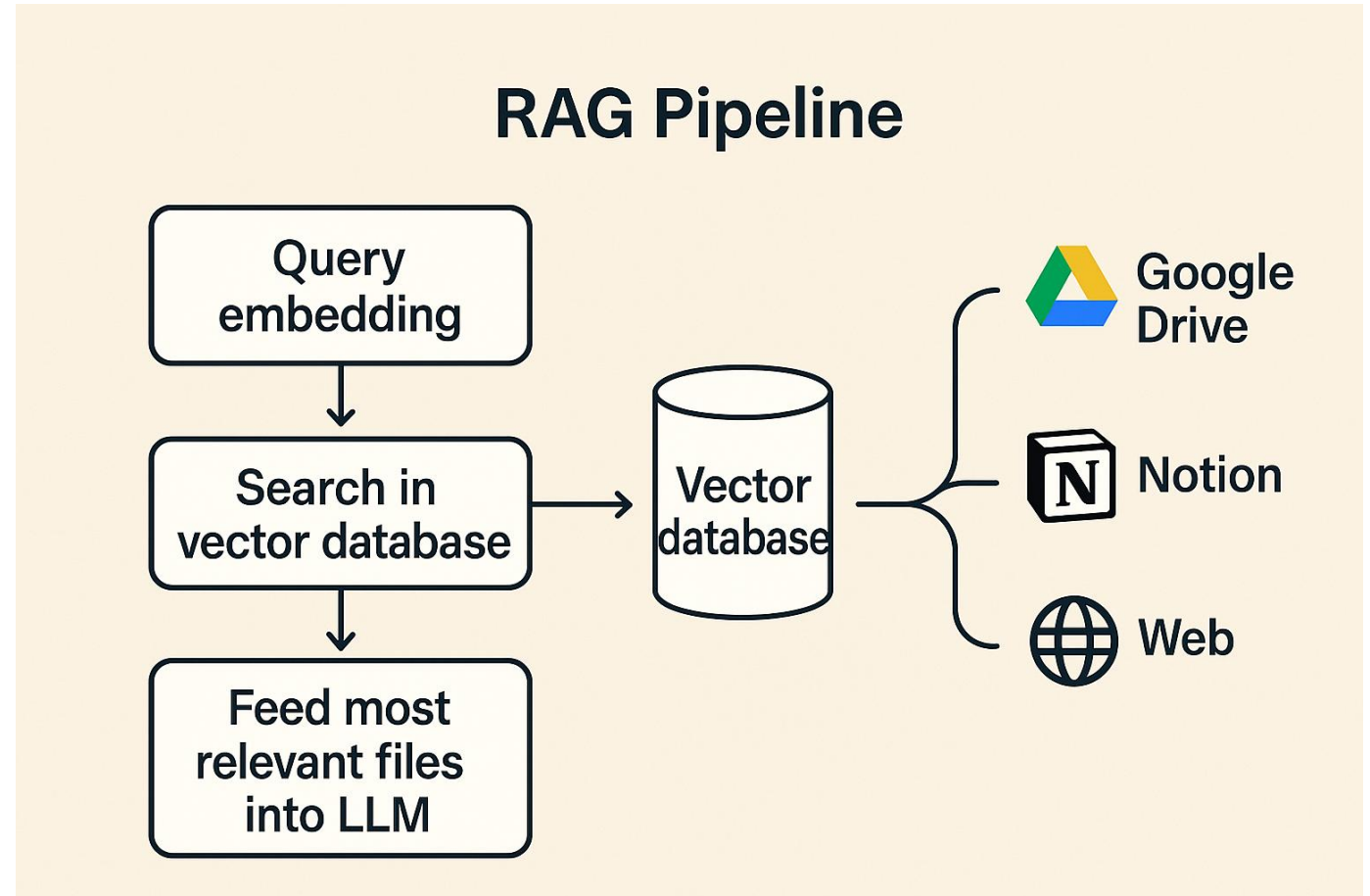


Retrieval-Augmented Generation (RAG)

Used when the answer depends on broader or historical documents

Pipeline :

- Query embedding
- Search in vector database
- Feed most relevant files into LLM
- Generates answer



When to use MCP vs RAG



MCP used when tools can be
executed fast

RAG used on mostly static and heavy
loaded data sources

Use Case	Method
Slack	MCP
Google Drive	RAG
Notion	RAG/MCP
Web	RAG

Where we are now

- Working prototype integrated in Slack
- Supports:
 1. Slack
 2. Google Drive
 3. Notion
 4. Web

Data is mocked for now due to access limitations, but full flow is operational

Working prototype integrated in Slack



Slack



Google Drive



Notion



Web



Mocked data

Demo Time

Reminder: small mock data for now



Demo

What's Next?

- Gmail integration:
Forward emails to the bot along with a query → bot reads context + use knowledge → replies with an answer
- Kubernetes deployment:
For always-on availability & internal hosting
- Additional sources:
Airtable? Or any of your suggestions

What's Next



Gmail integration

Forward emails to the bot along with a query
→ bot reads context + use knowledge → rep



Kubernetes deployment

For always-on availability & internal hosting



Additional sources

Airtable? Or any of your suggestions

We'd Love Your Feedback

- Which sources would you like the bot to include?
- Are there workflows you can think of that the bot could simplify?
- Any other suggestions/comments



Contact: eth.ai.assistant@gmail.com

We'd love your feedback

We're excited about the future of this project and would love your input

